

File Management

A guide to manage files in Jupyterlab user interface

Objectives

- Identify and name the file browser in JupyterLab and explain its UI elements.
- Perform general file management tasks

Menu Bar

File Edit View Run Kernel Git Tabs Settings Help

File Browser

File browser showing the file system/structure

Name	Last Modified
Boot_camp_test	a month ago
copy_bootcamp_sum...	18 days ago
datawhys-content-no...	a year ago
datawhys-intern-note...	a year ago
intro-jupyterlab	a month ago
intro-notebooks	a month ago
Test	a year ago
2022-06-06-intern-p...	a year ago
2022-06-21-intern-pr...	a year ago
boston_apartments.csv	3 months ago
DS_PS-1.ipynb	3 months ago
Untitled.ipynb	18 days ago

Launcher

Notebook

Python 3 R xpython

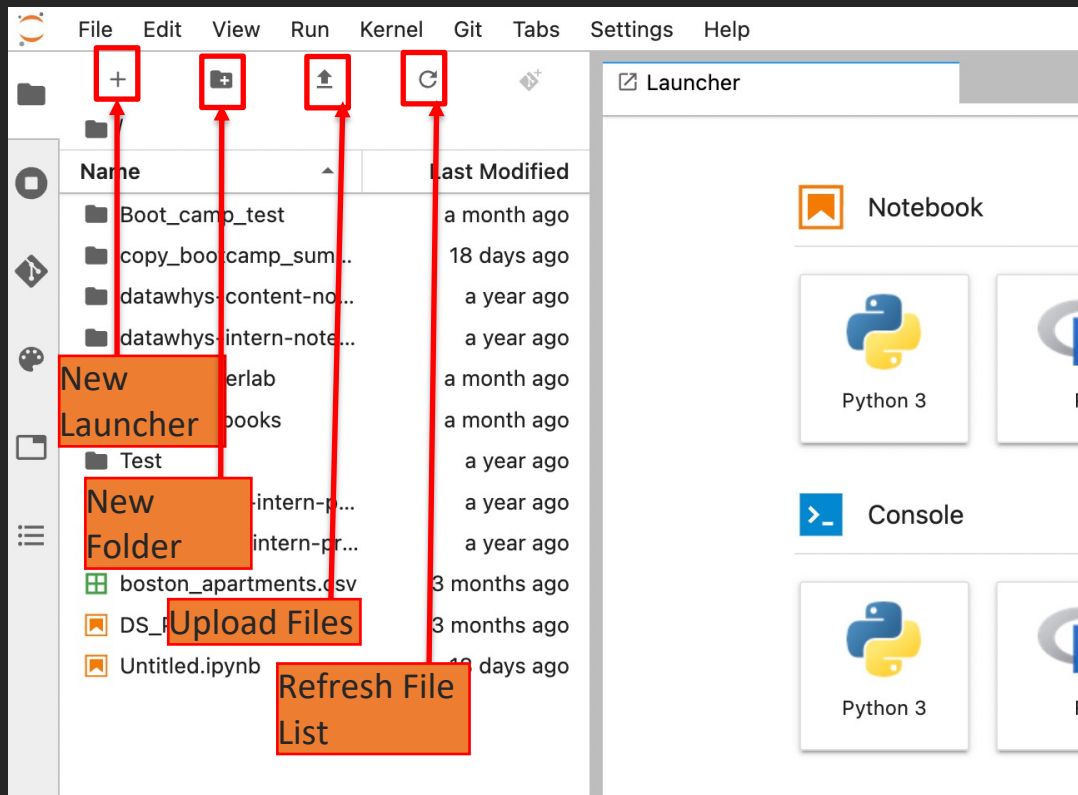
Console

Python 3 R xpython

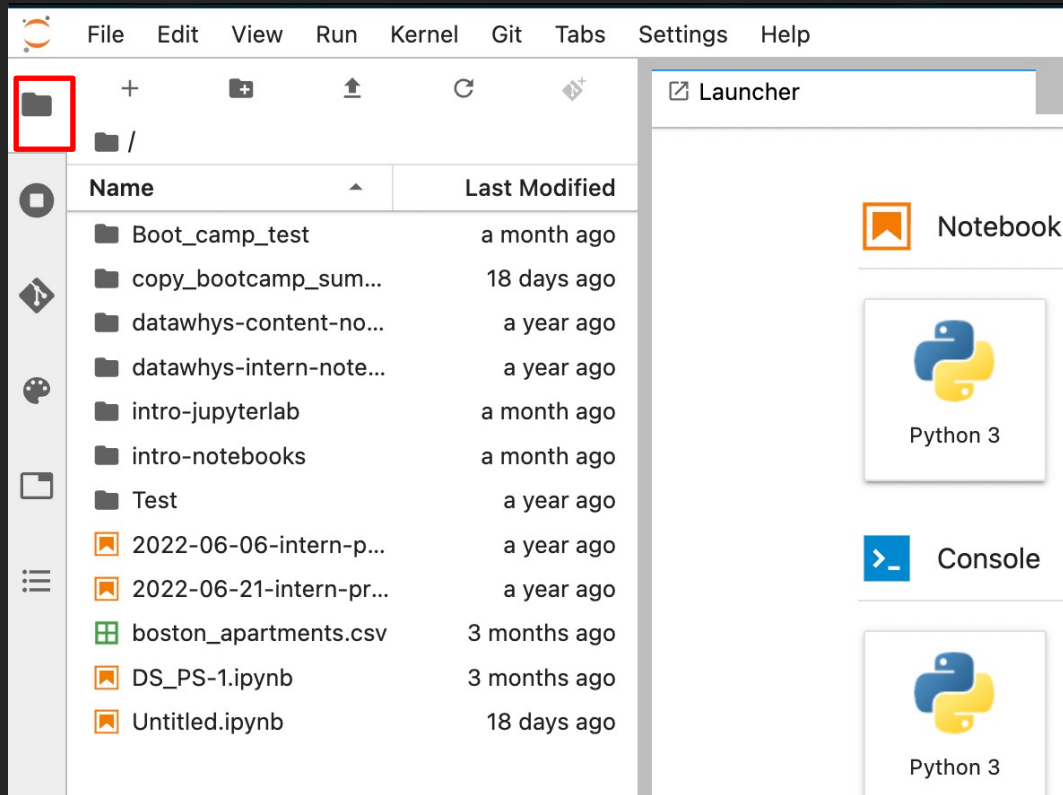
Other

Terminal Text File Markdown File Show Contextual Help

File Browser Details

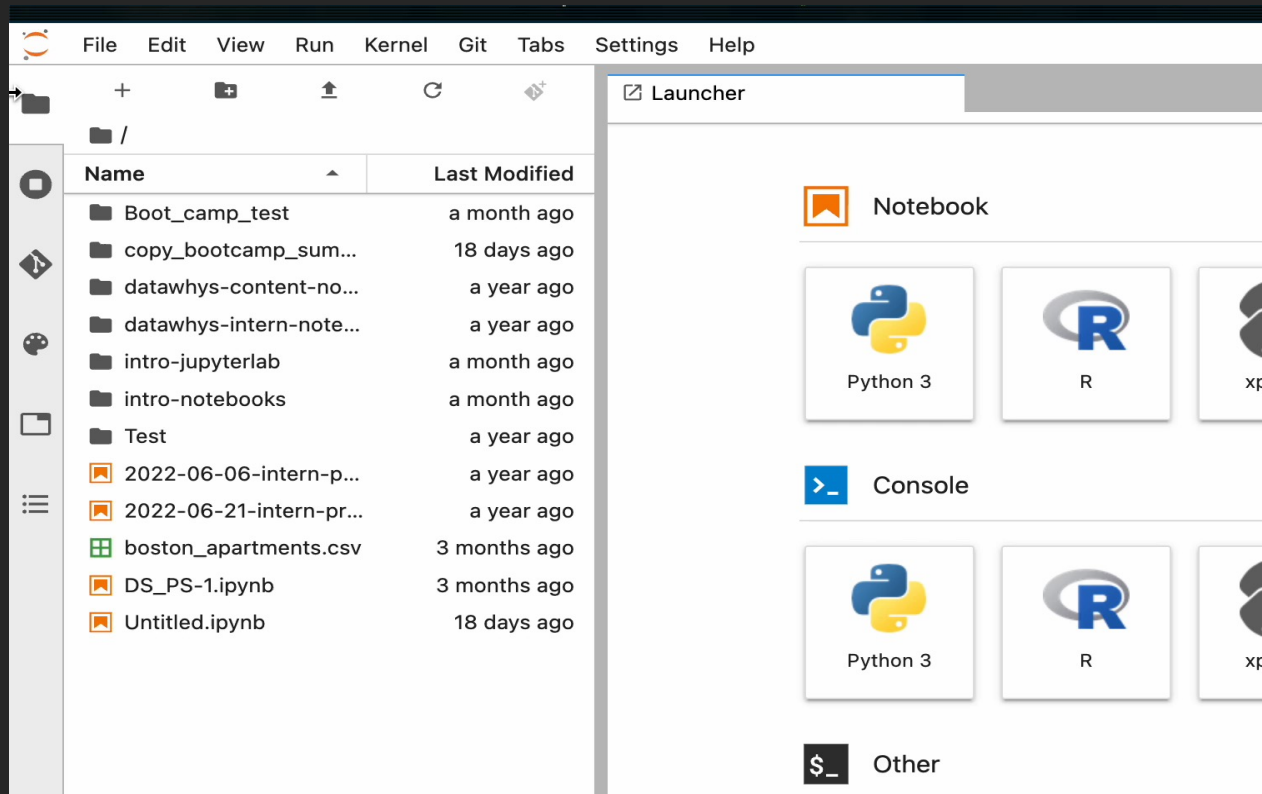


Hide/Unhide File Browser Details



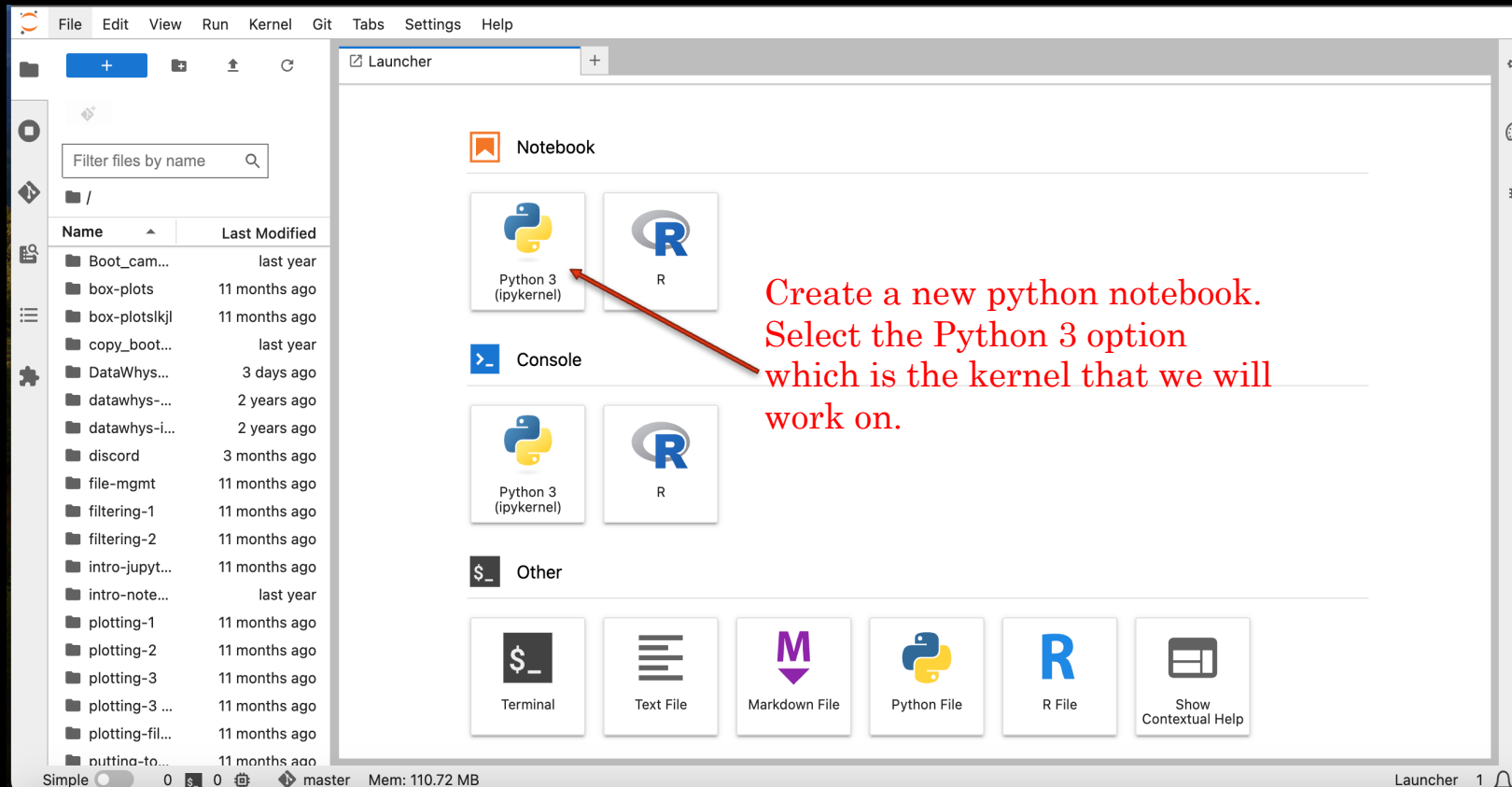
Click on the File Browser to hide/unhide the file browser details

Hide/Unhide File Browser Details



Click on the File Browser to hide/unhide the file browser details

Creating python file from launcher



The screenshot shows the JupyterLab Launcher interface. On the left is a file browser with a search bar and a list of files. The main area is titled 'Launcher' and contains three sections: 'Notebook', 'Console', and 'Other'. The 'Notebook' section has two options: 'Python 3 (ipykernel)' and 'R'. A red arrow points to the 'Python 3 (ipykernel)' option. To the right of the arrow, red text reads: 'Create a new python notebook. Select the Python 3 option which is the kernel that we will work on.'

File browser (left):

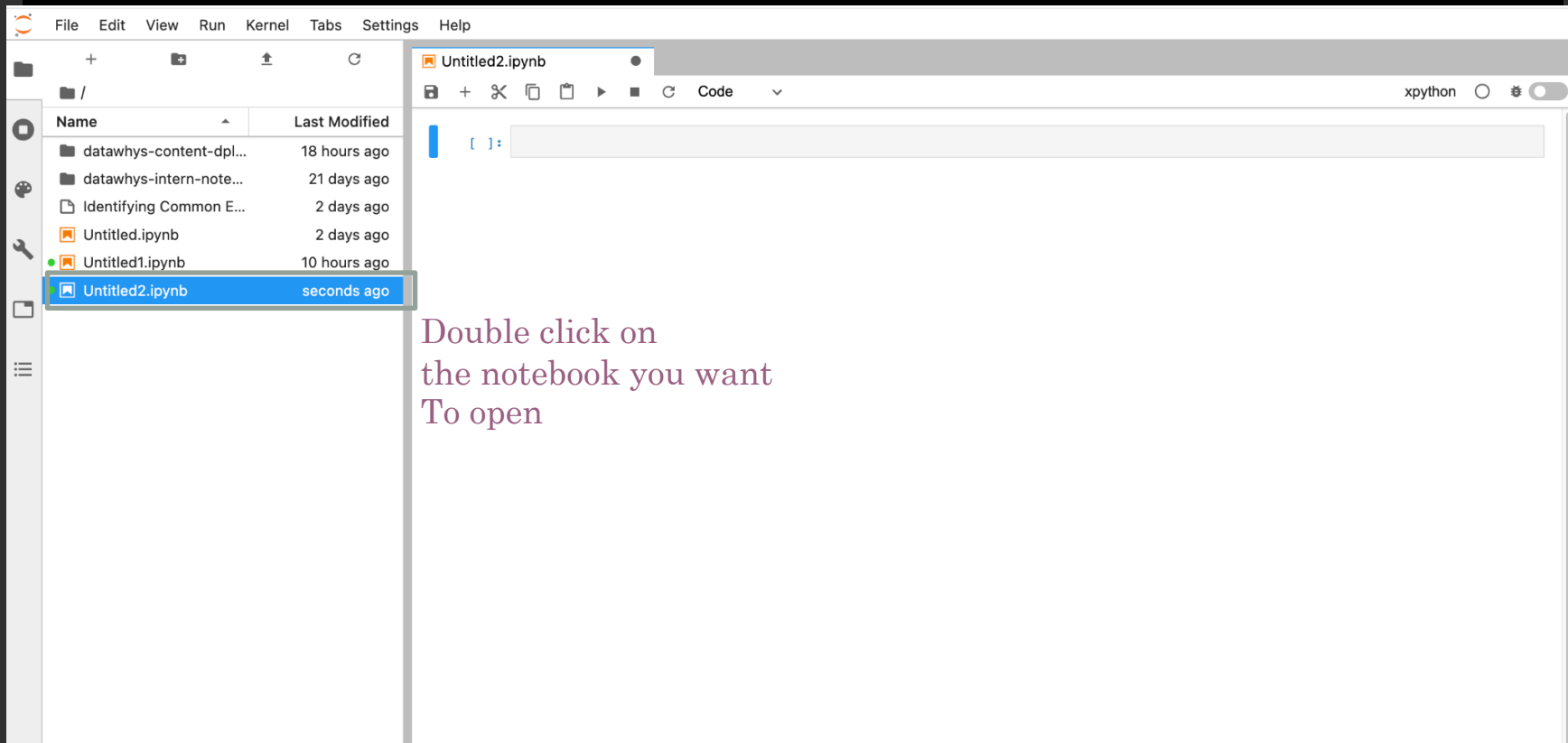
Name	Last Modified
Boot_cam...	last year
box-plots	11 months ago
box-plotsklj	11 months ago
copy_boot...	last year
DataWhys...	3 days ago
datawhys-...	2 years ago
datawhys-i...	2 years ago
discord	3 months ago
file-mgmt	11 months ago
filtering-1	11 months ago
filtering-2	11 months ago
intro-jupyt...	11 months ago
intro-note...	last year
plotting-1	11 months ago
plotting-2	11 months ago
plotting-3	11 months ago
plotting-3 ...	11 months ago
plotting-fil...	11 months ago
putting-to...	11 months ago

Launcher (main area):

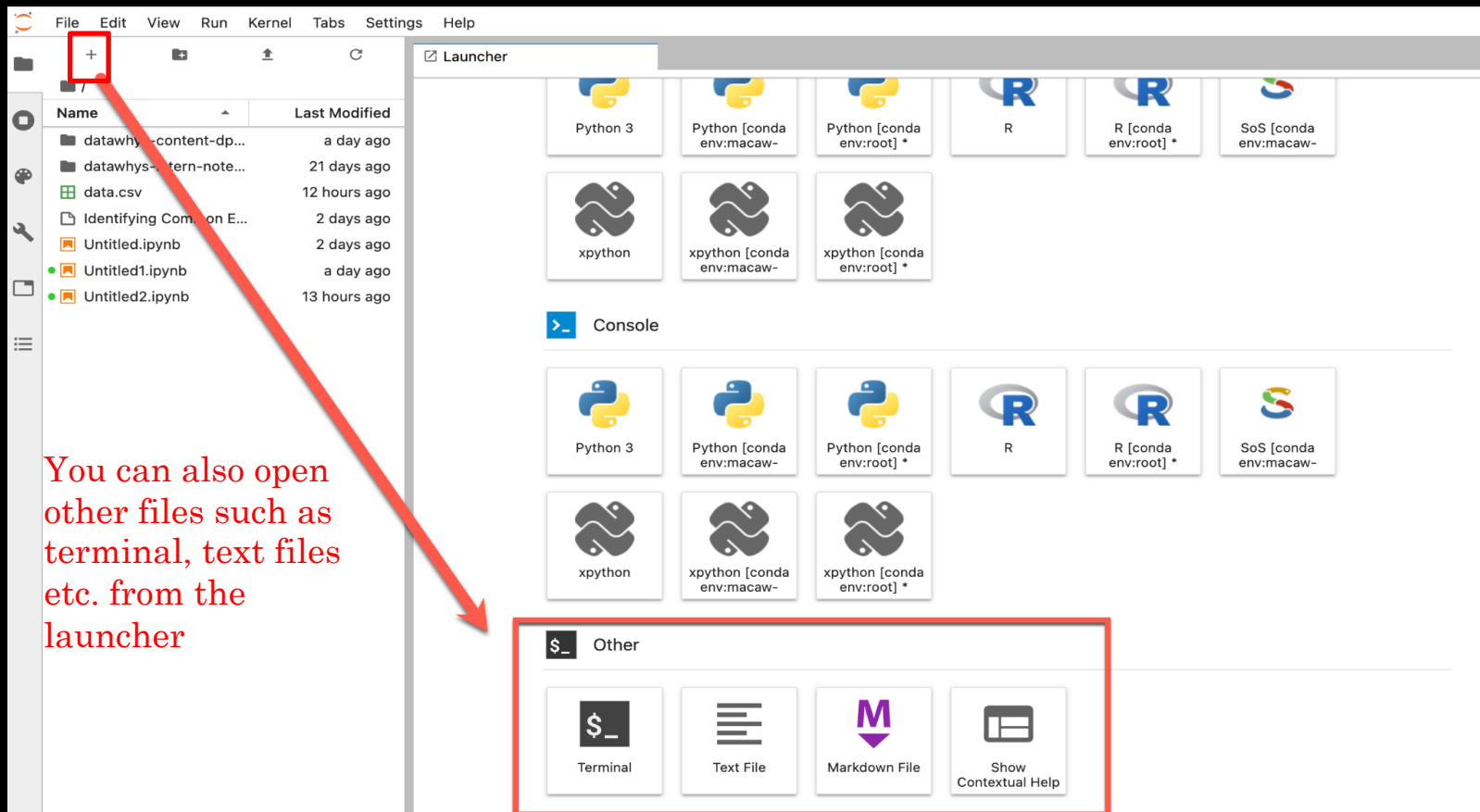
- Notebook**
 - Python 3 (ipykernel)
 - R
- Console**
 - Python 3 (ipykernel)
 - R
- Other**
 - Terminal
 - Text File
 - Markdown File
 - Python File
 - R File
 - Show Contextual Help

Status bar (bottom): Simple 0 \$ 0 master Mem: 110.72 MB Launcher 1

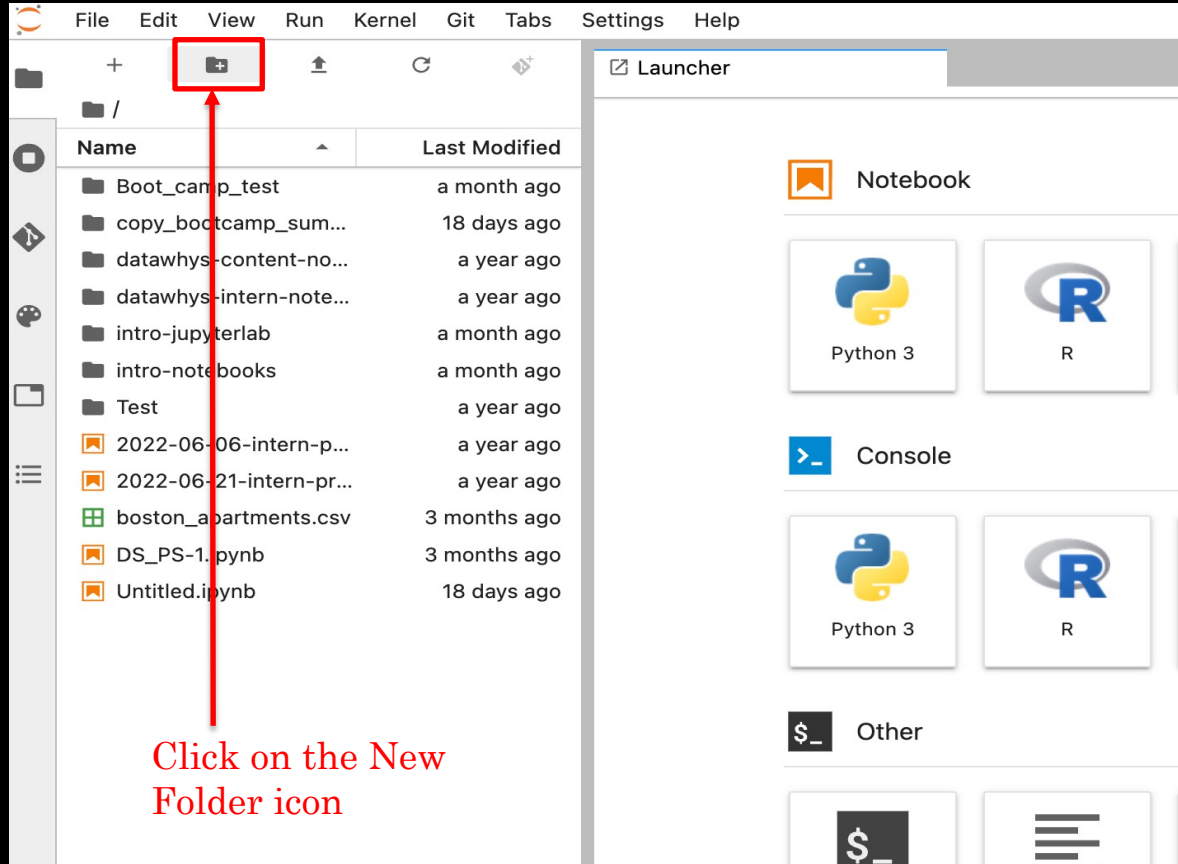
Open created python file



Opening other file types

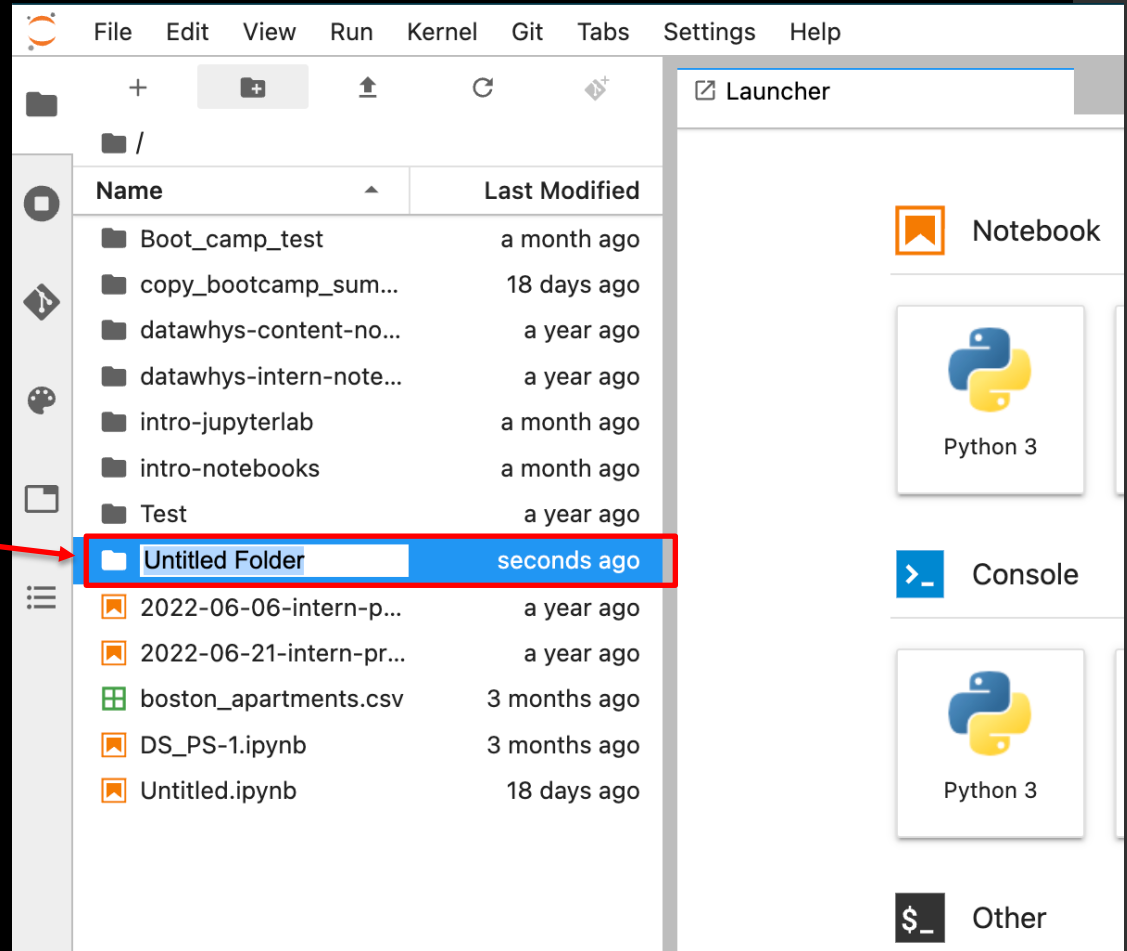


Creating a Folder



Creating a Folder

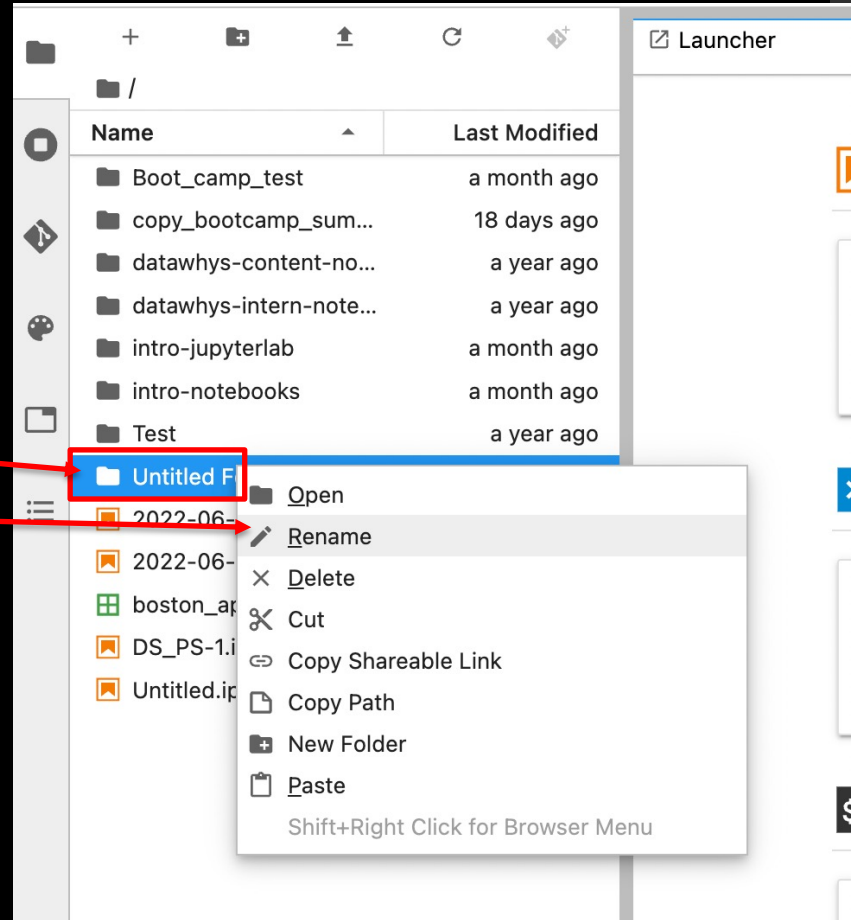
"Untitled Folder"
created



Renaming a Folder

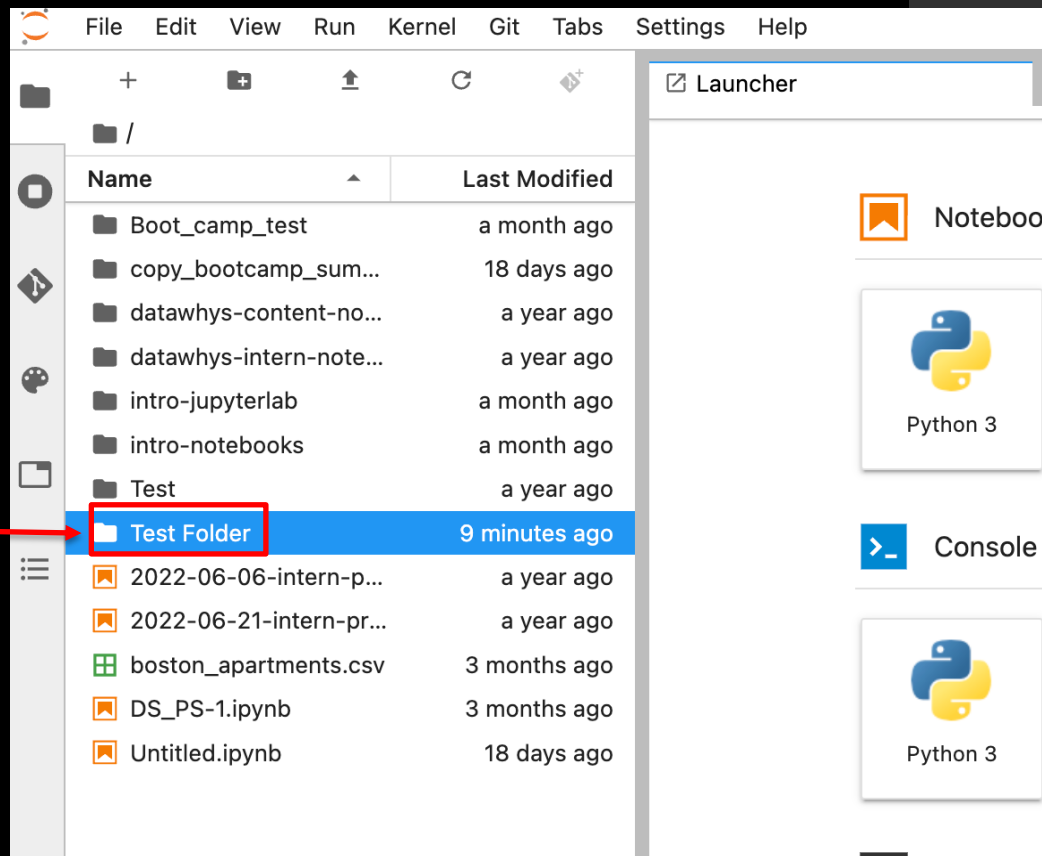
Right click on the
folder name

Click on Rename and
provide desired name
and press "Enter" or
Click anywhere



Renaming a Folder

Folder has been
renamed



Creating and Renaming a Folder

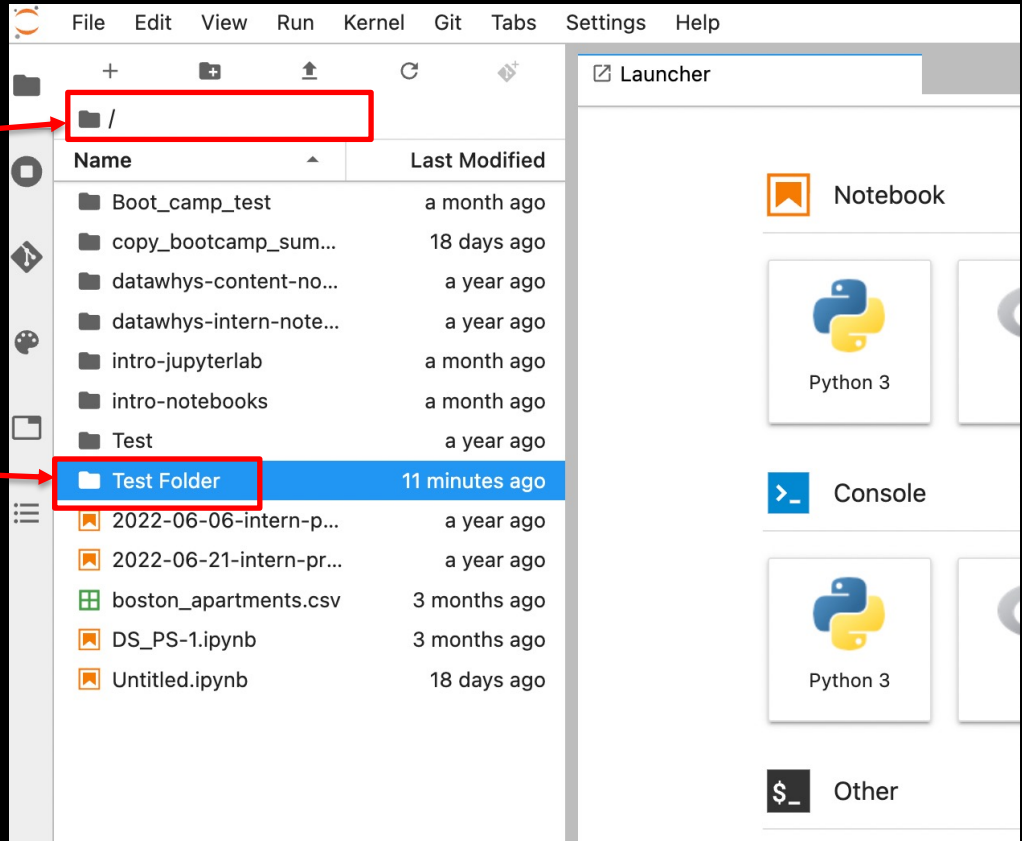
The screenshot displays the JupyterLab application window. The top menu bar includes File, Edit, View, Run, Kernel, Git, Tabs, Settings, and Help. Below the menu is a toolbar with icons for creating a new file, opening a recent file, uploading, refreshing, and deleting. The left sidebar contains a file browser view showing a directory structure. The main area is divided into two panes: a 'Launcher' pane on the right and a 'Console' pane below it. The Launcher pane shows a 'Notebook' section with icons for Python 3, R, and xpy. The Console pane shows a 'Console' section with the same icons. The file browser on the left lists files and folders with their last modified dates.

Name	Last Modified
Boot_camp_test	a month ago
copy_bootcamp_sum...	18 days ago
datawhys-content-no...	a year ago
datawhys-intern-note...	a year ago
intro-jupyterlab	a month ago
intro-notebooks	a month ago
Test	a year ago
2022-06-06-intern-p...	a year ago
2022-06-21-intern-pr...	a year ago
boston_apartments.csv	3 months ago
DS_PS-1.ipynb	3 months ago
Untitled.ipynb	18 days ago

Open Folder

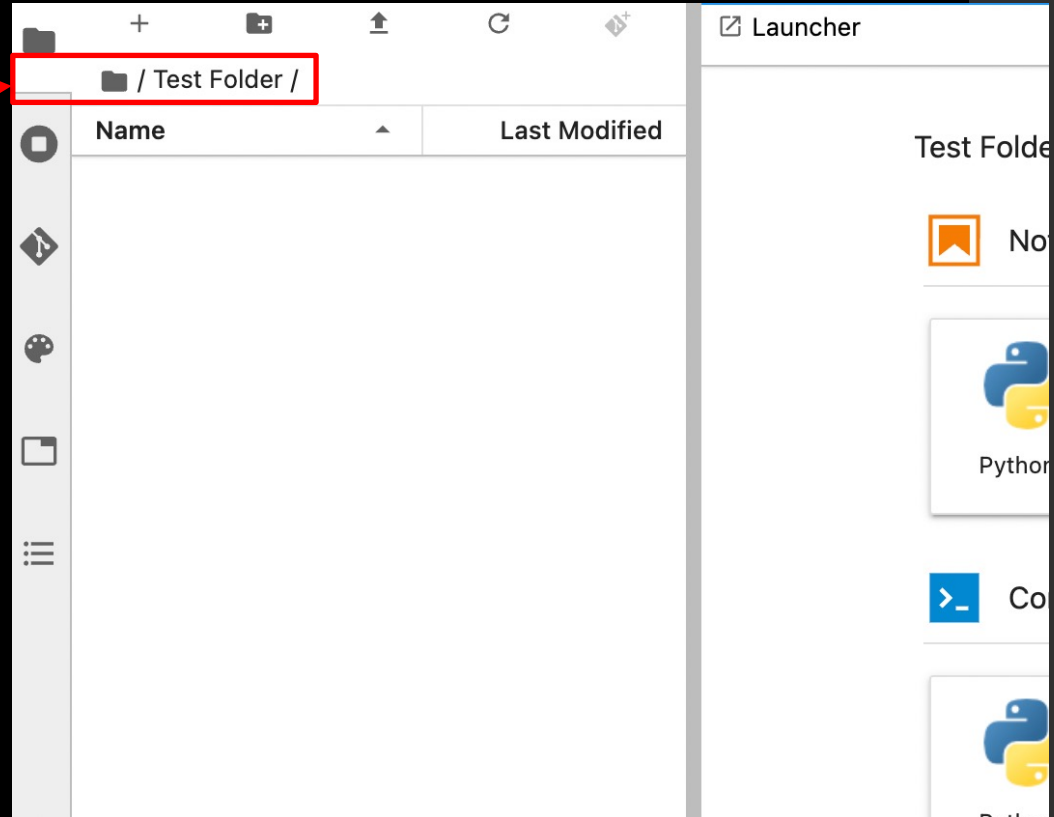
Before opening a new folder notice this.
This shows
our current location in
the file system.

Double click to open a
folder



Open Folder

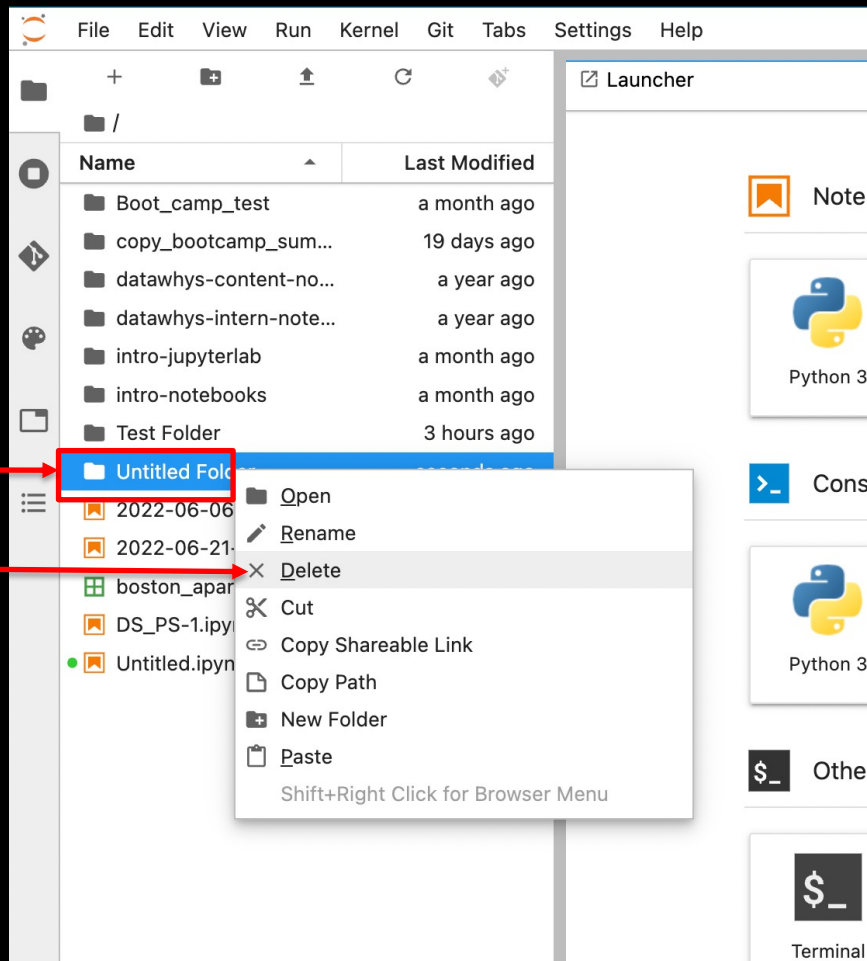
Indicates we are in
Test Folder



Deleting a Folder

Right click on the
folder name you want
to delete

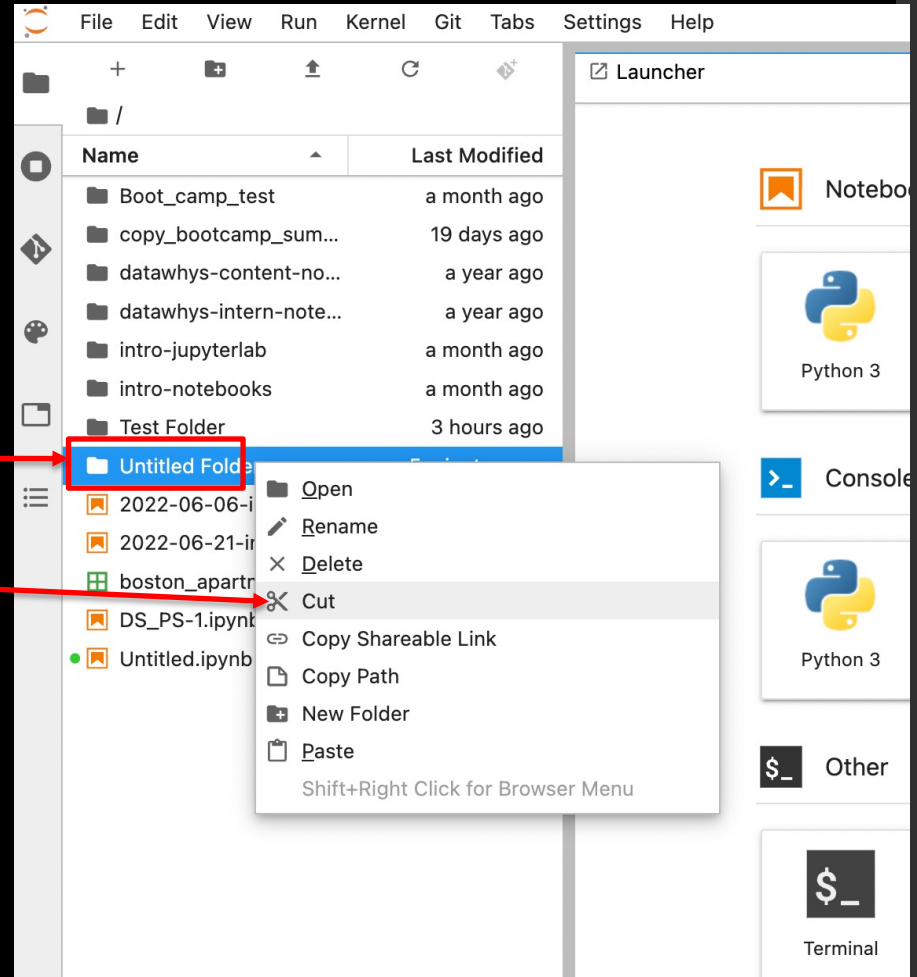
Click on Delete



Cut a Folder

Right click on the
folder name you want
to cut

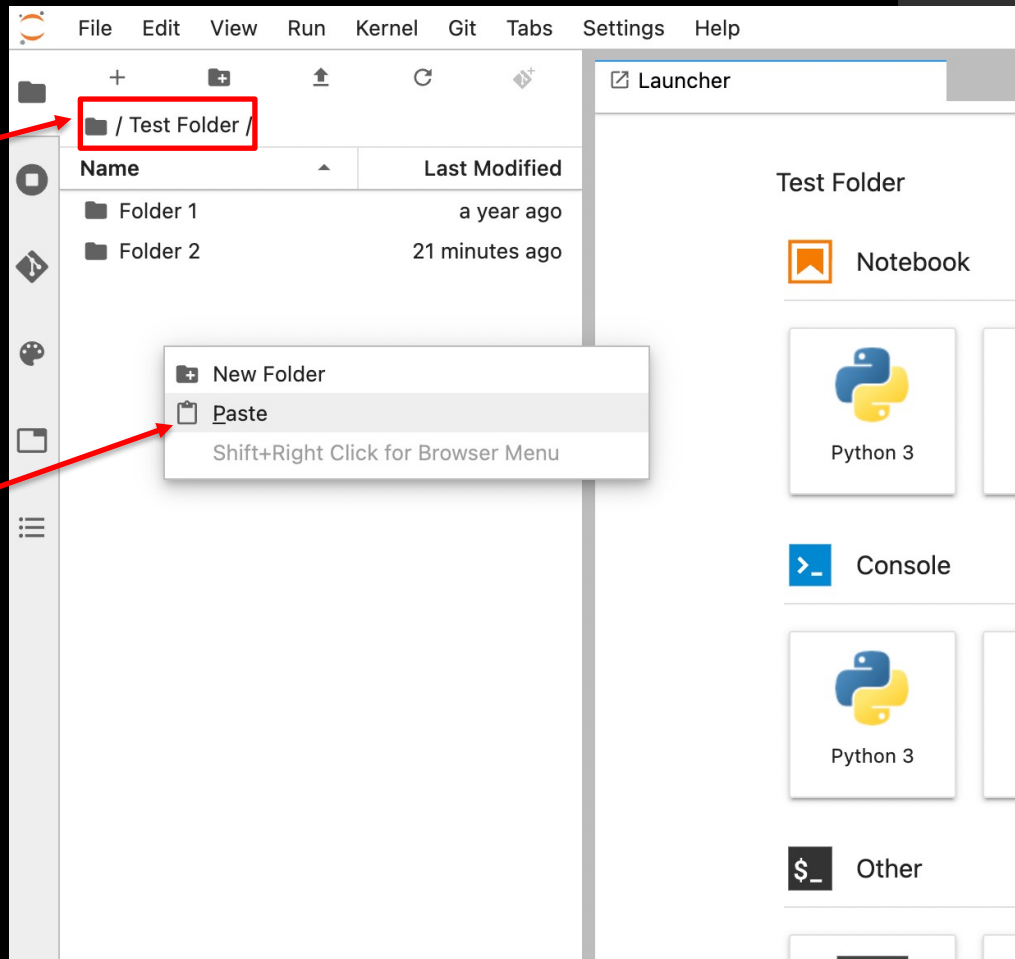
Click on Cut



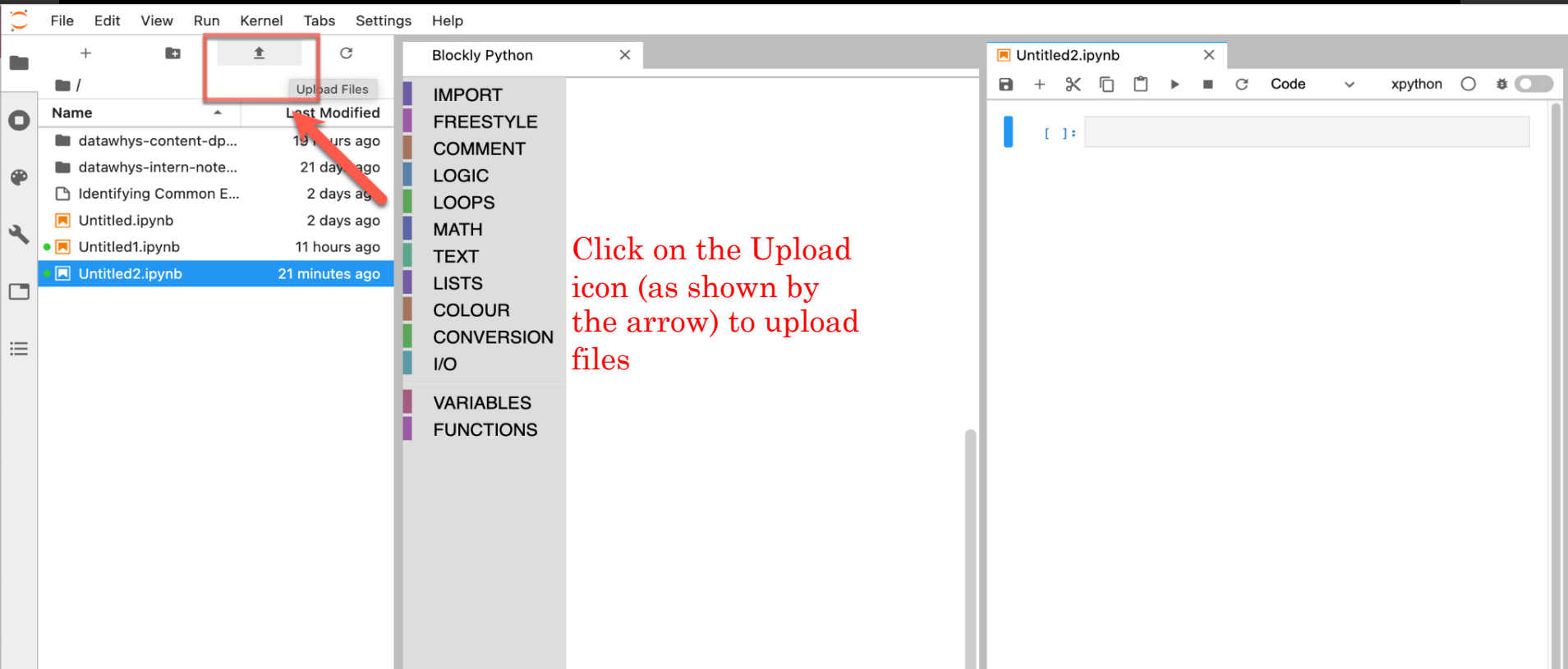
Cut a Folder

Open the folder where
you want to Paste

Right click and
select Paste



Uploading a file in Jupyterlab (1)



The screenshot displays the Jupyterlab web interface. At the top, a menu bar includes 'File', 'Edit', 'View', 'Run', 'Kernel', 'Tabs', 'Settings', and 'Help'. Below the menu, a toolbar contains icons for file operations. The 'File' menu is open, showing a list of files and folders. A red box highlights the 'Upload Files' button, which is represented by an upward-pointing arrow icon. A red arrow points from the text 'Click on the Upload icon (as shown by the arrow) to upload files' to the 'Upload Files' button. The file list shows several files, including 'Untitled2.ipynb' which is highlighted in blue. The right pane shows a code editor with a single line of code: `[ ]:`.

File Edit View Run Kernel Tabs Settings Help

Upload Files

Blockly Python

IMPORT
FREESTYLE
COMMENT
LOGIC
LOOPS
MATH
TEXT
LISTS
COLOUR
CONVERSION
I/O
VARIABLES
FUNCTIONS

Click on the Upload icon (as shown by the arrow) to upload files

Untitled2.ipynb

Code xpython

```
[ ]:
```

Uploading a file in Jupyterlab (2)

The screenshot displays the Jupyterlab web interface. On the left, the file browser shows a directory structure with a table of files. The file 'data.csv' is highlighted with a red rectangle. The central pane contains a 'Blockly Python' widget with a list of code blocks. The right-hand pane shows a notebook titled 'Untitled2.ipynb' with a code cell containing the text '[]:'.

Name	Last Modified
datawhys-content-dp...	19 hours ago
datawhys-intern-note...	21 days ago
data.csv	seconds ago
Identifying Common E...	2 days ago
Untitled.ipynb	2 days ago
Untitled1.ipynb	11 hours ago
Untitled2.ipynb	25 minutes ago

Blockly Python

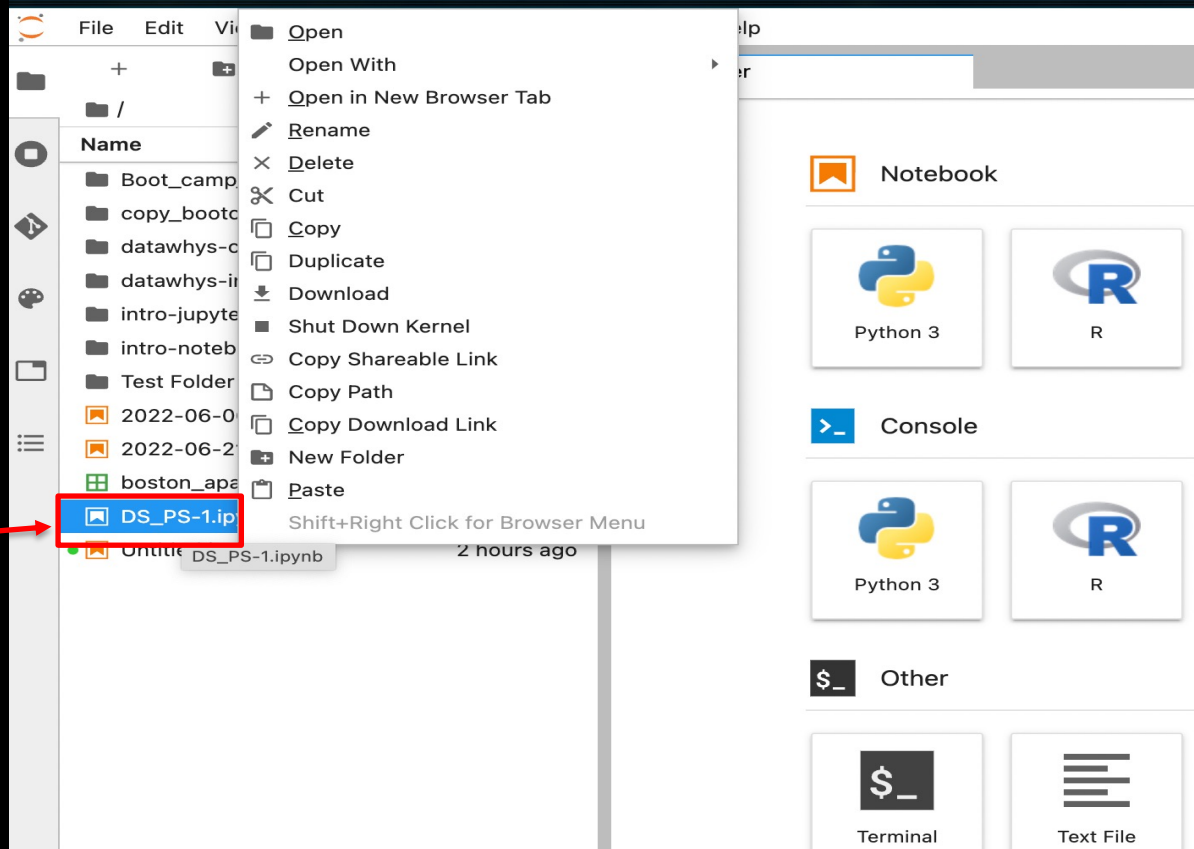
- IMPORT
- FREESTYLE
- COMMENT
- LOGIC
- LOOPS
- MATH
- TEXT
- LISTS
- COLOUR
- CONVERSION
- I/O
- VARIABLES
- FUNCTIONS

Untitled2.ipynb

[]:

File operations

Right Click on any file
to perform rename,
copy, cut, delete,
download etc.

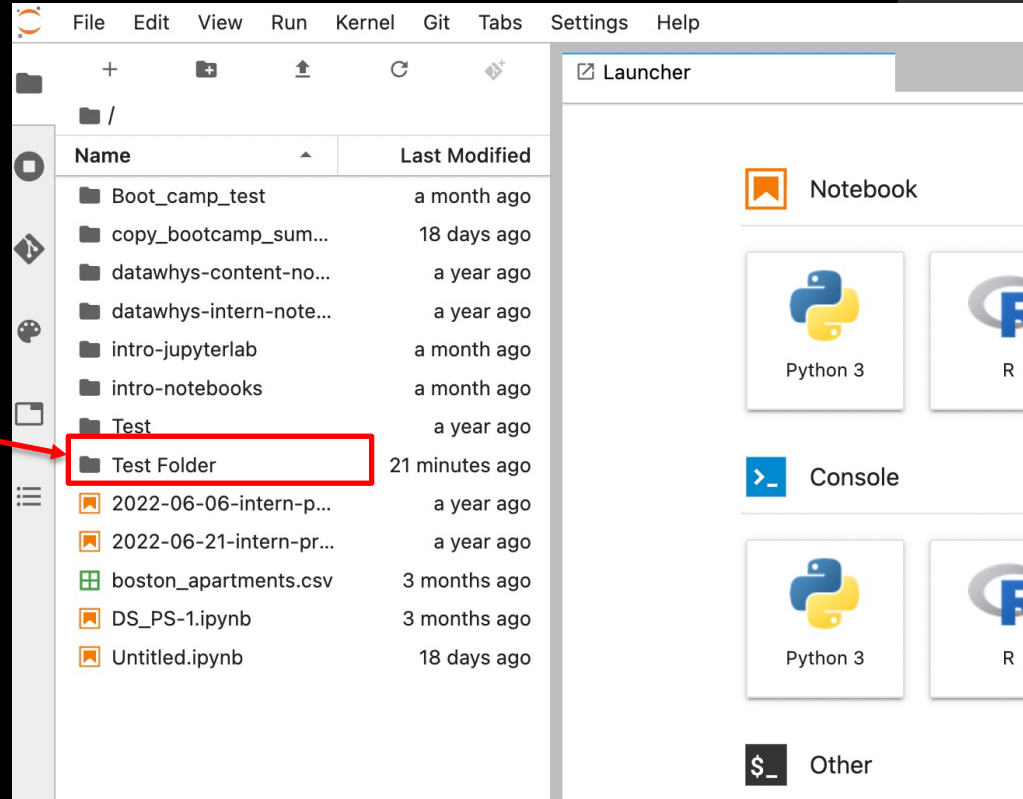


Upload a file in any specific folder

Suppose we want to upload a file in Test Folder. We've to

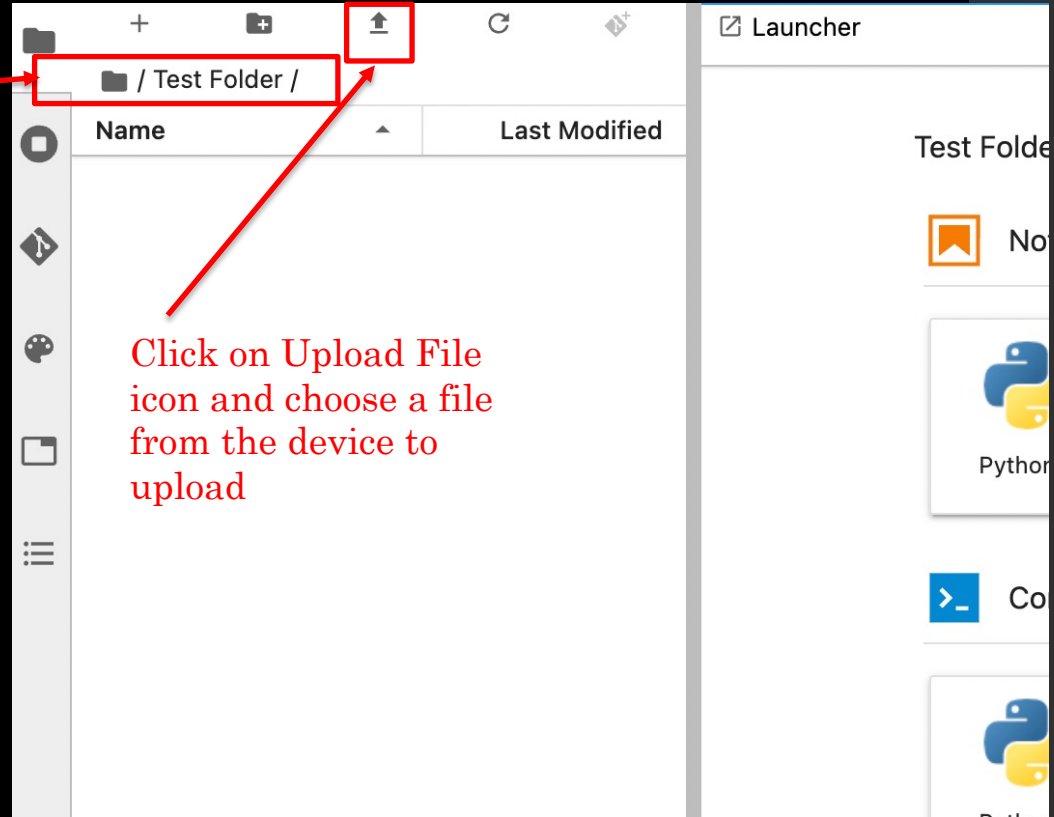
- Open Test folder
- Upload the file

Open Test Folder by
double clicking



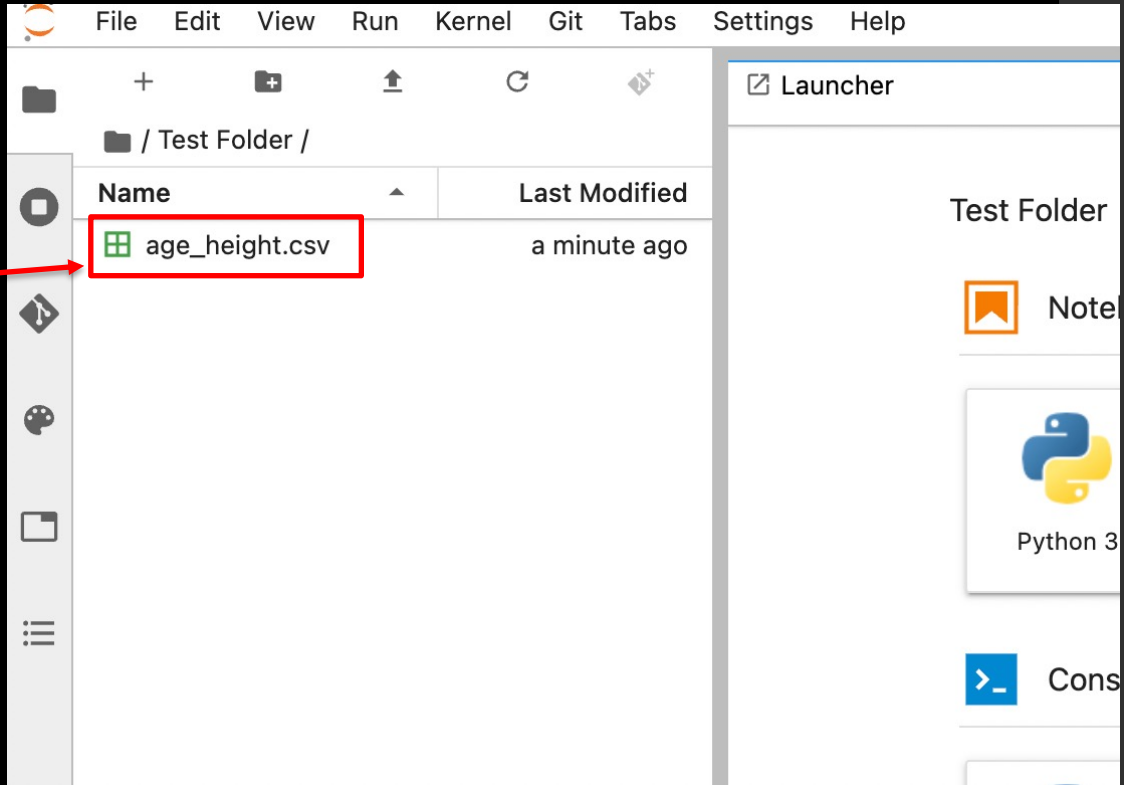
Upload a file in any specific folder

Indicates we are in
Test Folder



Upload a file in any specific folder

A file has been
uploaded



File Path

- A file path specifies the location of a file in a file system/structure.
- They can be used to locate files.
- Example: `"/home/username/Documents/File1.txt"`.

In file system each folder is also known as "directory" and in each directory are separated with separator `"/"`

File Path

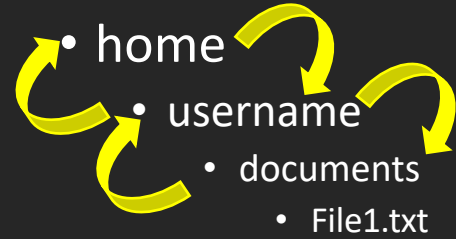
Path: `"/home/username/Documents/File1.txt"`.

- This path indicates that in our "home" directory or folder we have "username" folder.
- Within the "username" folder we have "Documents" folder.
- Within the "Documents" folder we have our targeted file "File.txt".

- home
 - username
 - documents
 - File1.txt

File Path

Path: `"/home/username/Documents/File1.txt"`.



- "home" is the **parent directory** of "username"
- "username" is the **parent directory** of "documents"

Alternatively, we can say

- "username" is the **child directory** of "home"
- "documents" is the **child directory** of "username"

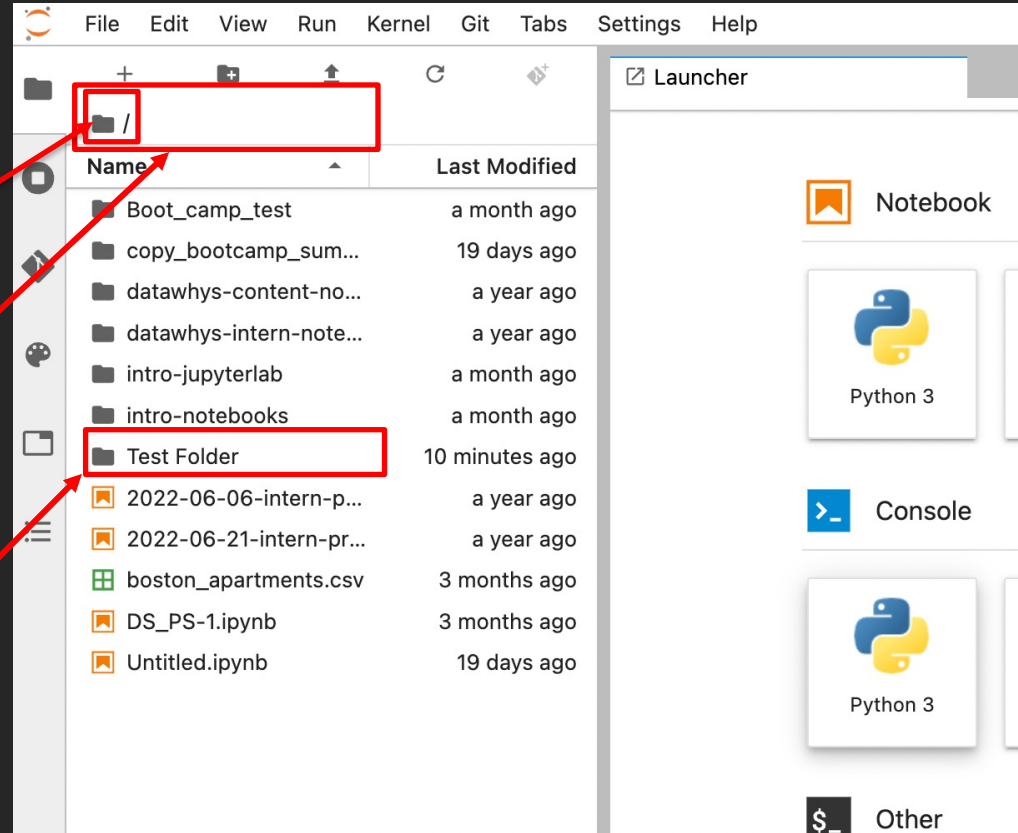
File Path in Jupyter Lab

- After Login the interface shows like this

This "folder/" icon indicates root folder/directory for each user

In JupyterLab current directory is shown in this area

Let's open this folder and see

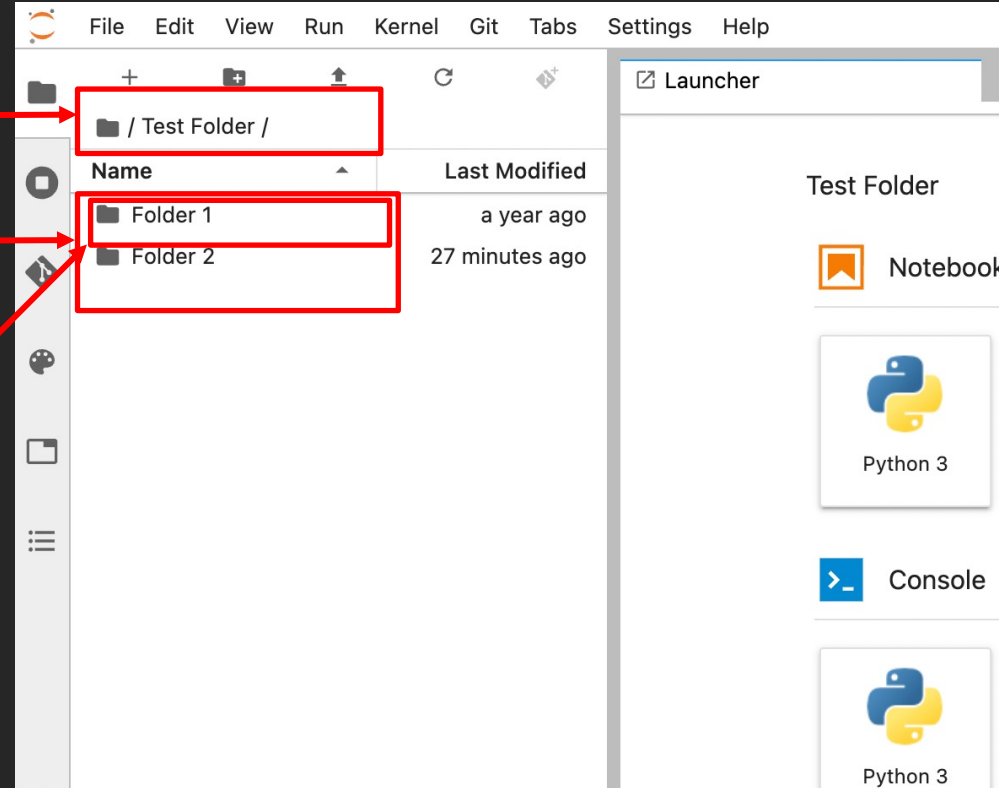


File Path in Jupyter Lab

Indicates that we are within Test Folder and current directory is Test Folder

Shows the file system of Test Folder

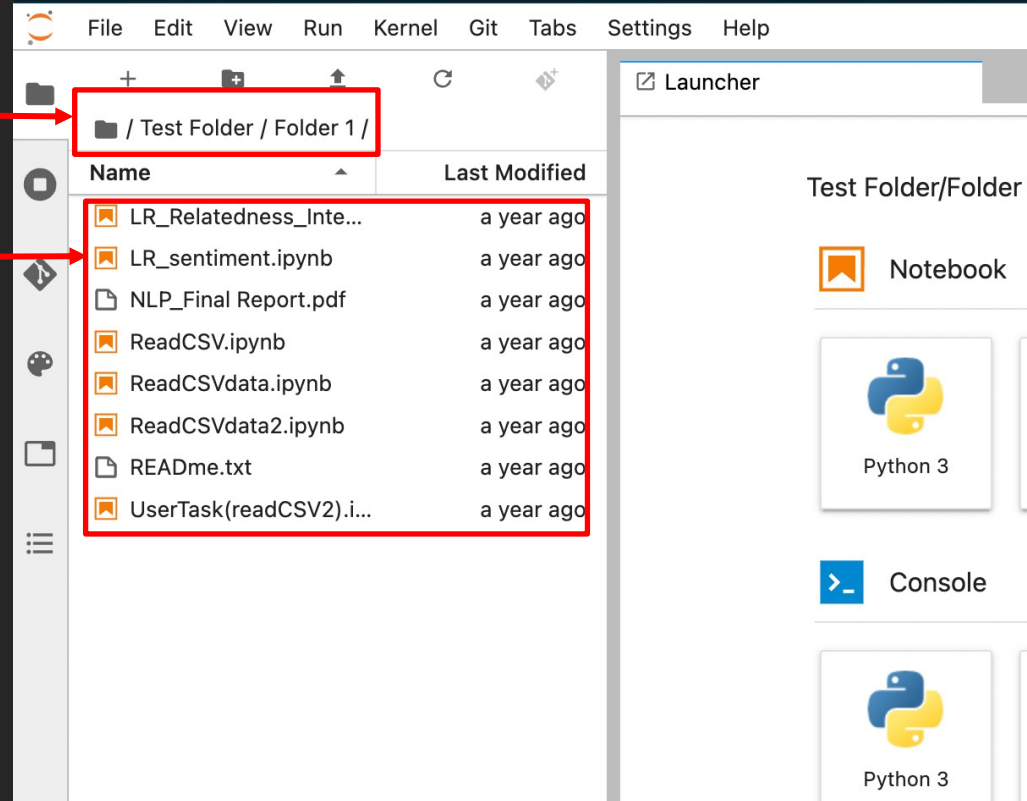
Let's open Folder 1 and see



File Path in Jupyter Lab

Indicates that we are within Folder 1
and current directory is Folder 1

Shows the
files within Folder 1



Types of File Path

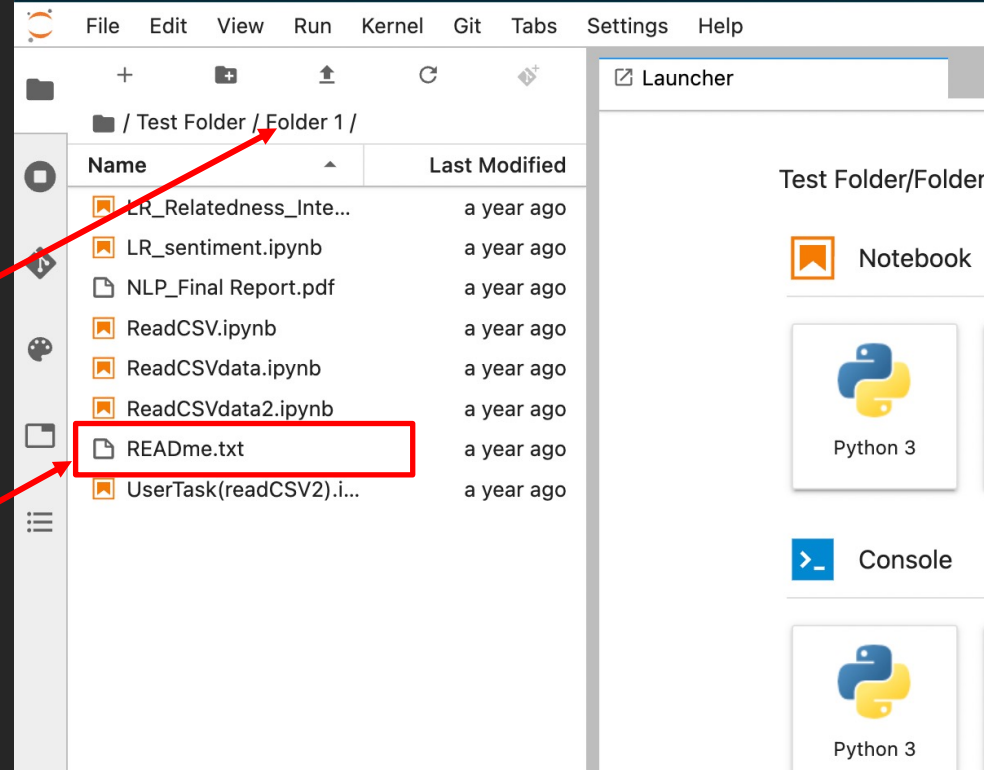
- Relative Path
- Absolute Path

Relative Path

- A relative path specifies the location of a file or folder relative to the current working directory.

Current directory is Folder 1

README.txt file is located
in current directory.
The relative path of this file is
"README.txt"



Relative Path

- Suppose we have this file structure under our root directory. And our **current directory is Folder 1**.
- What is the relative path of Universities.csv?

Ans: "Universities.csv"

- Test Folder
 - Folder 1
 - **Universities.csv**
 - Folder 3
 - Students.csv
 - Folder 2
 - Teachers.csv



Given file structure

Relative Path

- Suppose we have this file structure under our root directory. And our **current directory is Folder 1**.
- What is the relative path of Students.csv?

- Test Folder
 - Folder 1
 - Universities.csv
 - Folder 3
 - **Students.csv**
 - Folder 2
 - Teachers.csv

Relative Path

- Suppose we have this file structure under our root directory. And our **current directory is Folder 1**.
- What is the relative path of Students.csv?
 - Notice that Students.csv is under Folder 3
 - Folder 3 is the **child directory** of Folder 1
 - The relative path of this Students.csv will be **"Folder 3/Students.csv"**

By this we
go to Folder 3
as Folder 3 is
child directory
of Folder 1

- Test Folder
 - Folder 1
 - Universities.csv
 - Folder 3
 - Students.csv
 - Folder 2
 - Teachers.csv

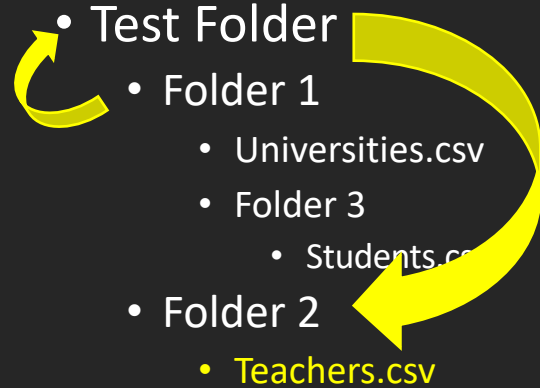
Relative Path

- Suppose we have this file structure under our root directory. And our **current directory is Folder 1**.
- What is the relative path of Teachers.csv?

- Test Folder
 - Folder 1
 - Universities.csv
 - Folder 3
 - Students.csv
 - Folder 2
 - **Teachers.csv**

Relative Path

- Suppose we have this file structure under our root directory. And our **current directory is Folder 1**.
- What is the relative path of Teachers.csv?
- To navigate to Teachers.csv
 - First ,we have to go the parent directory Folder 1 which is Test Folder. For this we use "../"
 - Then we go to Folder 2 which is a child directory of Test Folder. For this we use "Folder2/"
 - Path will be **"../Folder2/Teachers.csv"**



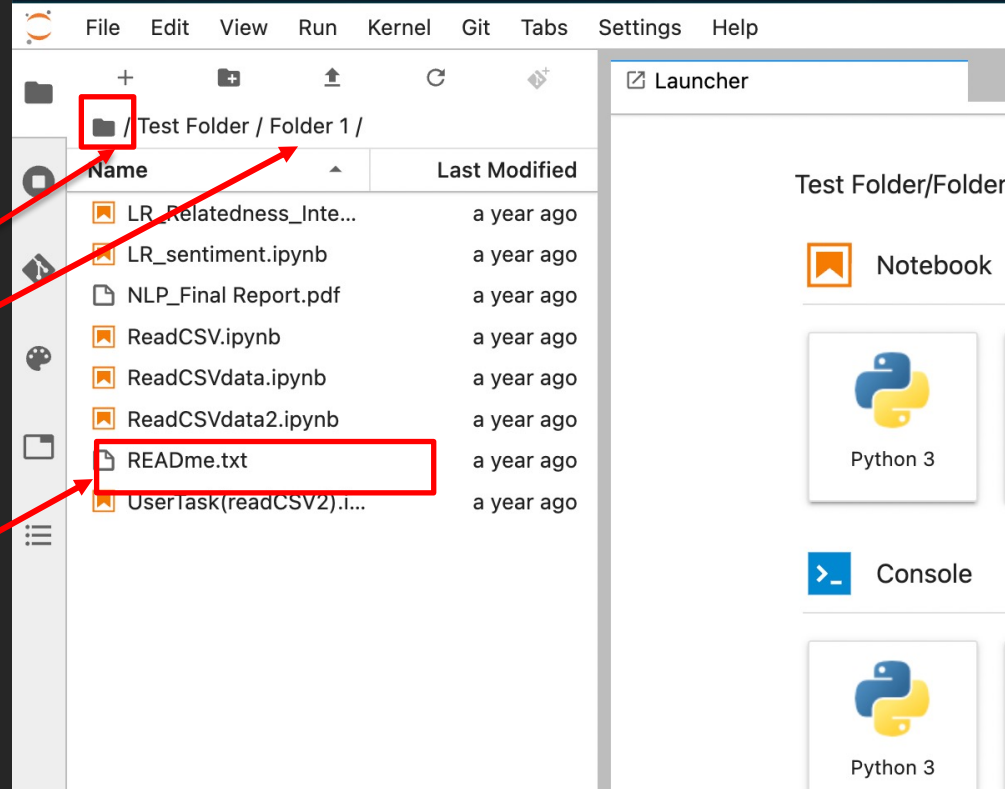
Absolute Path

- An absolute path provides the complete location of a file or folder starting **from the root directory**

This "folder/" icon indicates root directory for each user

Current directory is Folder 1

The absolute path of this file is "Test Folder/READme.txt"



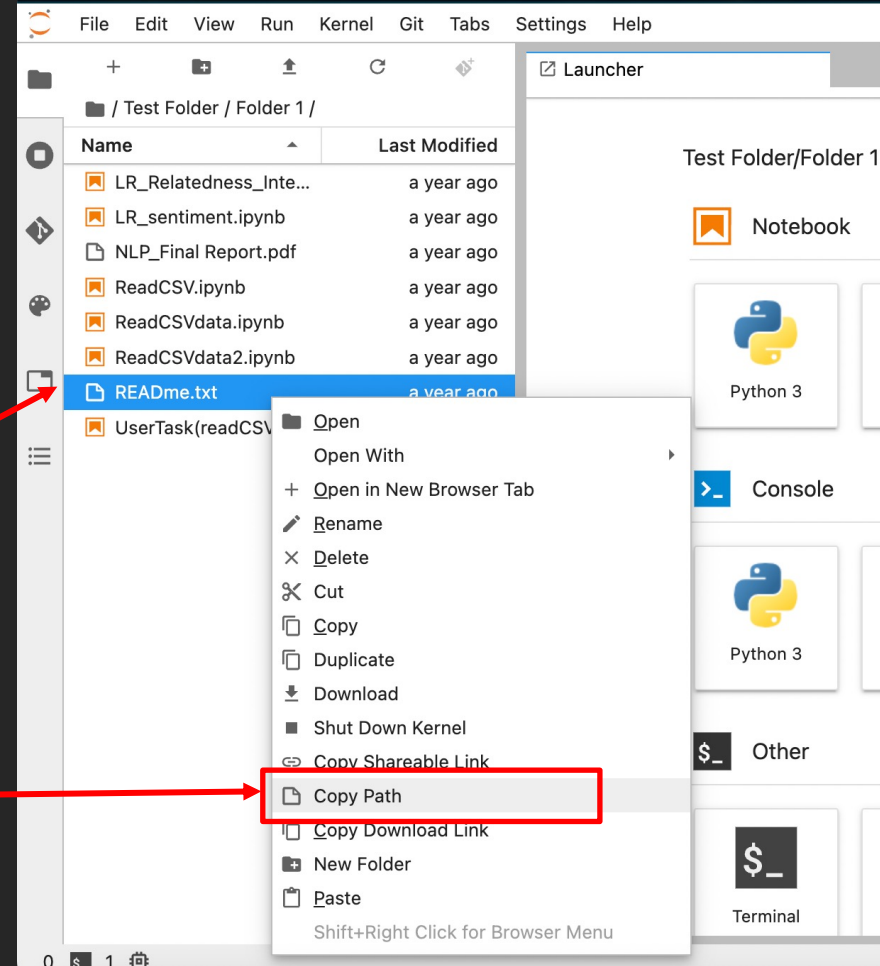
- An absolute path doesn't depend on current directory

Absolute Path in JupyterLab

- In JupyterLab absolute path can be obtained by "Copy Path"

Right click on a file

Clicking on "Copy Path" will copy the absolute path in the clipboard. You can paste this in any text



Absolute Path

Suppose we have this file structure under our root directory.

- What is the absolute path of Students.csv if our **current directory is Folder 1**?
- What is the absolute path of Students.csv if our **current directory is Folder 2**?

- Test Folder
 - Folder 1
 - Universities.csv
 - Folder 3
 - Students.csv
 - Folder 2
 - Teachers.csv

Absolute Path

Suppose we have this file structure under our root directory.

- What is the absolute path of Students.csv if our **current directory is Folder 1**?
- What is the absolute path of Students.csv if our **current directory is Folder 2**?

- Test Folder
 - Folder 1
 - Universities.csv
 - Folder 3
 - Students.csv
 - Folder 2
 - Teachers.csv

Absolute path doesn't depend on current directory. For both cases they are the same which is "Test Folder/Folder3/Students.csv"

Activity

- Open the Activity document and proceed with the tasks